Niloofar Kazemi

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Summary

Machine learning researcher building deep learning methods for medical imaging and other data-scarce settings. Experience across vision, language technologies and scientific modeling, with strong Python and PyTorch skills.

Research Experience

ML Research Assistant at *McGill University Health Centre (MUHC)* *Since Feb. 2026*
– Research Institute of MUHC

- Achieved 76% AUC in two-year sarcoma metastasis prediction under data-scarce conditions (51 samples) using a Multi-Instance Contrastive Learning pipeline.
- Achieved a 99% AUC for locating tumors in sarcoma patients' MRI scans by fine-tuning a RadImageNet-pretrained ResNet-50.
- Implemented an MRI preprocessing workflow featuring N4 bias field correction, fat/muscle-based intensity normalization, and spatial resampling.

Research Assistant at *University of Alberta* *Sep. 2023 – Dec. 2025*
– Unsworth Lab

- Performed two elastin-like polypeptide (ELPs) characterization through X-ray photoelectron spectroscopy, contact angle, and Fourier transform infrared spectroscopy (FTIR), and ELP ellipsometry
- Investigated the temperature transition effect of two ELPs on their hydration and rehydration using quartz crystal microbalance with dissipation (QCM-D)
- Analyzed uremic toxin-protein layer adsorption behavior over time using QCM-D

Research Intern at *Sholeh Sanat Manufacturing Company* *Jul. 2021 – Oct. 2021*
– Department of Simulation and Optimization

- Designed deterministic models to simulate industrial processes and component interactions
- Optimized system constraints and continuous variables, ensuring strict alignment with real-world target metrics and operational data

Publications and Conferences

Kazemi, N., Vallières, M., Gorelik, N. Tsui, J. 2026. Predicting soft-tissue sarcoma metastasis risk from MRI under data scarcity: A patch-based deep attention multiple-instance learning approach. In preparation for *RSNA's Radiology: Artificial Intelligence (RadiologyAI)*. Also accepted at the *WiML@NeurIPS'26* workshop.

Shayegh, B., **Kazemi, N.** 2026. Mind which bird you favour: Parameterizing adequacy–fluency balance in meta-evaluation of machine translation. Published in *Proceedings of the Conference on Machine Translation (WMT)*. Independent Research

Kazemi, N., Ghahremanzadeh, A., Unsworth, L. 2024. Effect of indoxyl sulfate on albumin adsorbed amounts and layer rigidity. Presented at *the Annual Alberta Biomedical Engineering Conference*. Oral (top 20%)

Education

M.Sc. at *University of Alberta* *2023 – 2026*
– Major in Biomedical Engineering
– Thesis. End-tethered elastin-like polypeptides: Phase behaviour and indoxyl sulphate Affinity.

B.Sc. at *Amirkabir University of Technology* *2018 – 2022*
– Major in Chemical Engineering
– Thesis. Design of carbon capture process from a cement production plant.

Service and Teaching

Mentorship and Teaching Assistantship

Research Mentor at *University of Alberta*
– Trained a high school student *(Summer 2025)* and an undergraduate student *(Spring 2024)* in lab techniques, buffer and uremic toxin solution preparation, and QCM-D experimentation

TA at *University of Alberta*
– Introductory University Chemistry II *(Winter 2025)*
– Calculus for Engineering I & II *(Winter, Fall 2024)*
– Engineering Thermodynamics *(Fall 2024)*

TA at *Farzanegan 6 High School*
– Mathematics and Calculus

Volunteer Work

- **Event Coordinator** for Science to Business Network (S2BN), 2024 – 2025.
- **Event Coordinator** for Chem-E-Car Nomination Annual Competition, 2020.

Notable Course Projects

Breast cancer classification from gene expression data at
– Classified microarray mRNA expression data into normal and breast cancer subtypes, leveraging machine learning techniques for multiclass classification.

Targeting Prototheca with Tubulin Binding Agents at
– Developed an alternative therapeutic strategy for mastitis in dairy cows by identifying antimicrotubular agents targeting Prototheca Zopfii tubulin, using homology modeling, molecular docking, and dynamics simulations.

Skills

Programming	Python (Pandas, Numpy, PyTorch, Matplotlib), C/C++, MATLAB
Tools	LaTeX, Jupyter Notebook, Molecular Operating Env., Visual Molecular Dynamics, CasaXPS, Aspen HYSYS, DWSIM - Open Source Chemical Process Simulator, QCMD Acoustic Wave Sensors (AWSensors)